



Access S3 from a VPC



Pelin Ece Burgun

```
[ec2-user@ip-10-0-3-33 ~]$ aws s3 ls s3://nextwork-vpc-project-pelin
2026-08-15 12:07:13      2431554 NextWork - Denzel is awesome.png
2026-08-15 12:07:10      2399812 NextWork - Lelo is awesome.png
[ec2-user@ip-10-0-3-33 ~]$ sudo touch /tmp/test.txt
[ec2-user@ip-10-0-3-33 ~]$ aws s3 cp /tmp/test.txt s3://nextwork-vpc-project-pelin
upload: ../../tmp/test.txt to s3://nextwork-vpc-project-pelin/test.txt
[ec2-user@ip-10-0-3-33 ~]$ aws s3 ls s3://nextwork-vpc-project-pelin
2026-08-15 12:07:13      2431554 NextWork - Denzel is awesome.png
2026-08-15 12:07:10      2399812 NextWork - Lelo is awesome.png
2026-08-15 12:17:43           0 test.txt
[ec2-user@ip-10-0-3-33 ~]$
```



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC (Virtual Private Cloud) is a logically isolated virtual network dedicated to you within the AWS cloud. It allows you to configure your IP address range, subnets, and gateways, enabling you to run your resources (like EC2 instances) in a secure, fully controlled network environment. It is the foundation of security and traffic routing in cloud architecture.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to create a securely isolated network environment. I launched an EC2 instance inside the VPC's public subnet and used it as a secure base to interact with Amazon S3, demonstrating how resources inside a VPC can communicate with AWS services that live outside of it.

One thing I didn't expect in this project was...

One thing I didn't expect was that an EC2 instance doesn't automatically have the permissions to interact with other AWS services like S3. I learned that even though you are logged into the AWS console, the instance itself needs to be explicitly authenticated using IAM Access Keys (or ideally, an IAM Role) to use the AWS CLI.



This project took me...

This project took me about 90 minutes to complete. This included the time spent setting up the VPC infrastructure from scratch, generating the access keys, and running the CLI commands in the terminal to upload and view the files.



In the first part of my project...

Step 1 - Architecture set up

In this step, we are building the foundational network infrastructure for the project. We are creating a new Virtual Private Cloud (VPC) from scratch and then launching a new EC2 instance into the public subnet of this VPC.

Step 2 - Connect to my EC2 instance

We are connecting directly to our newly launched EC2 instance via the browser using EC2 Instance Connect, which grants us access to its command line (terminal)

Step 3 - Set up access keys

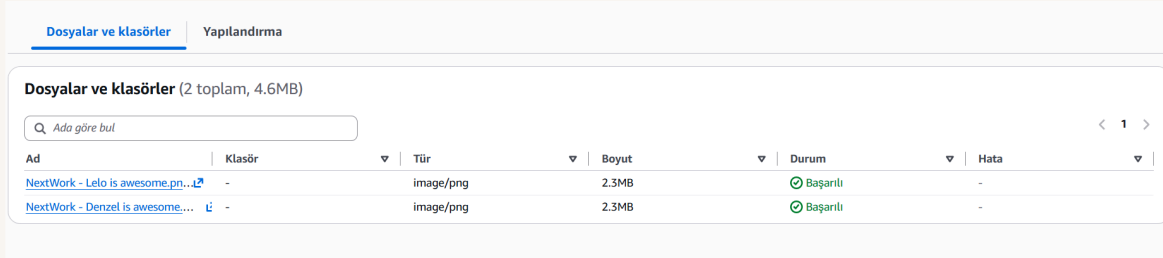
We are generating the credentials our EC2 instance needs to securely access AWS services. We are creating a new set of Access Keys through the IAM console specifically for CLI use.



Architecture set up

I launched one VPC (with a 10.0.0.0/16 CIDR block and 1 Public Subnet) and one Amazon Linux 2023 t2.micro EC2 instance running in its public subnet. I also created a new Security Group to allow SSH access to this instance.

I set up a new General purpose S3 Bucket (e.g., nextwork-vpc-project-pelin) and uploaded two files from my computer so that I can access and interact with them from my EC2 instance later.



The screenshot shows the AWS S3 console interface. At the top, there are two tabs: 'Dosyalar ve klasörler' (selected) and 'Yapılandırma'. Below the tabs, the text 'Dosyalar ve klasörler (2 toplam, 4.6MB)' is displayed. A search bar with the placeholder 'Ada göre bul' is present. Below the search bar is a table with the following columns: Ad, Klasör, Tür, Boyut, Durum, and Hata. The table contains two rows of data, both representing PNG images that were uploaded successfully.

Ad	Klasör	Tür	Boyut	Durum	Hata
NextWork - Lelo is awesome.png	-	image/png	2.3MB	Başarılı	-
NextWork - Denzel is awesome.png	-	image/png	2.3MB	Başarılı	-



Running CLI commands

The AWS Command Line Interface (CLI) is a tool that allows you to control and manage AWS services directly from the command line. Engineers use the CLI alongside scripts to automate the management of AWS resources and speed up operations.

The first command I ran was `aws s3 ls`. This command is used to list all the S3 buckets in the AWS account. However, it initially failed because the EC2 instance didn't have the necessary credentials set up yet.

The second command I ran was `aws configure`. This command is used to set up your AWS CLI by prompting you to enter your authentication details (like your Access Key ID, Secret Access Key, and Default Region).

```
[ec2-user@ip-10-0-3-33 ~]$ aws s3 ls
Unable to locate credentials. You can configure credentials by running "aws login".
[ec2-user@ip-10-0-3-33 ~]$ aws configure
AWS Access Key ID [None]:
```



Access keys

Credentials

The `aws configure` command is used to set up the authentication credentials (Access Key ID, Secret Access Key, AWS Region, and default output format) that the AWS CLI needs to function properly.

Access keys are credentials that allow applications, servers (like an EC2 instance), or the AWS CLI to securely authenticate and interact with AWS services programmatically. They consist of two parts: an Access Key ID (which acts like a username) and a Secret Access Key (which acts like a password).

The secret access key acts as a password that pairs with your Access Key ID (your username). Both are required to access AWS services. Because it is highly sensitive, it should never be shared with anyone and must be stored securely.

Best practice

The recommended best practice, rather than manually generating and copying access keys, is to create an IAM Role with the necessary S3 access permissions and attach that role directly to the EC2 instance.



In the second part of my project...

Step 4 - Set up an S3 bucket

We are creating a new storage bucket in Amazon S3 and uploading two sample files from our local computer into this bucket.

Step 5 - Connecting to my S3 bucket

We are returning to the EC2 terminal to configure the IAM access keys we just downloaded. Then, using AWS CLI commands, we view the contents of our S3 bucket and copy a newly created file from the EC2 terminal into S3.



Connecting to my S3 bucket

The first command I ran was `aws s3 ls`. This command is used to list all the S3 buckets in the AWS account. However, it initially failed because the EC2 instance didn't have the necessary credentials set up yet.

The terminal responded with a list of the existing S3 buckets in my account. I was able to spot my newly created bucket, `nextwork-vpc-project-pelin`, in this list.

```
[ec2-user@ip-10-0-3-33 ~]$  
[ec2-user@ip-10-0-3-33 ~]$ aws s3 ls  
2026-08-15 12:05:02 nextwork-vpc-project-pelin  
[ec2-user@ip-10-0-3-33 ~]$
```



Connecting to my S3 bucket

To list the specific objects inside the bucket, I ran the following command: `aws s3 ls s3://nextwork-vpc-project-pelin`

```
[ec2-user@ip-10-0-3-33 ~]$ aws s3 ls s3://nextwork-vpc-project-pelin
2026-08-15 12:07:13      2431554 NextWork - Denzel is awesome.png
2026-08-15 12:07:10      2399812 NextWork - Lelo is awesome.png
[ec2-user@ip-10-0-3-33 ~]$
```



Uploading objects to S3

First, I ran `sudo touch /tmp/test.txt` to create an empty text file in the EC2 instance.

The second command in that sequence was `aws s3 cp /tmp/test.txt s3://nextwork-vpc-project-pelin`. This command (cp stands for copy) is used to take the empty text file we just created on the EC2 instance and upload it directly to our S3 bucket.

The third command was `aws s3 ls s3://nextwork-vpc-project-pelin`. I ran this command to list the contents of my specific bucket one more time. This allowed me to verify that the test.txt file was successfully uploaded and now lives inside the S3 bucket!

```
[ec2-user@ip-10-0-3-33 ~]$ aws s3 ls s3://nextwork-vpc-project-pelin
2026-08-15 12:07:13      2431554 NextWork - Denzel is awesome.png
2026-08-15 12:07:10      2399812 NextWork - Lelo is awesome.png
[ec2-user@ip-10-0-3-33 ~]$ sudo touch /tmp/test.txt
[ec2-user@ip-10-0-3-33 ~]$ aws s3 cp /tmp/test.txt s3://nextwork-vpc-project-pelin
upload: ../../tmp/test.txt to s3://nextwork-vpc-project-pelin/test.txt
[ec2-user@ip-10-0-3-33 ~]$ aws s3 ls s3://nextwork-vpc-project-pelin
2026-08-15 12:07:13      2431554 NextWork - Denzel is awesome.png
2026-08-15 12:07:10      2399812 NextWork - Lelo is awesome.png
2026-08-15 12:17:43           0 test.txt
[ec2-user@ip-10-0-3-33 ~]$
```



nextwork.ai

The place to learn & showcase your skills

Check out nextwork.ai for more projects

